

PROJECT REPORT

THREAT INTELLIGENCE AGGREGATOR (NON-AI)

A Project Report Submitted for
Unified Mentor Internship
Project

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ABSTRACT

In today's rapidly evolving digital environment, cyber threats have become one of the major challenges for organizations and security teams. Attackers continuously use malicious IP addresses, domains, URLs, file hashes, and other indicators to perform activities such as malware distribution, phishing, data theft, and unauthorized access. Traditional security monitoring methods often require manual collection and analysis of threat information from multiple sources, which can be time-consuming and may delay threat detection. To address these challenges, this project focuses on developing a Threat Intelligence Aggregator (Non-AI) that helps automate the collection, processing, and analysis of cyber threat information.

This project implements a rule-based threat intelligence processing system using Python programming and Open-Source Intelligence (OSINT) feeds. The system collects threat indicators from multiple external sources and processes them through different modules including IOC extraction, validation, normalization, correlation, severity classification, and block list generation. Unlike Artificial Intelligence or Machine Learning-based solutions, this system relies on predefined security rules and logical analysis to identify and prioritize potential threats.

The system extracts different types of Indicators of Compromise (IOCs) such as IP addresses, URLs, domains, and file hashes from raw threat feeds. These indicators are validated and converted into a standard format to improve accuracy and usability. The correlation engine analyzes repeated indicators across multiple threat sources and assigns severity levels based on their occurrence. High-risk indicators are automatically added to generated block lists that can be used by security controls such as firewalls, endpoint protection systems, and web filters.

The project provides a practical approach for improving Security Operations Center (SOC) workflows by reducing manual investigation efforts and providing structured threat intelligence reports. It demonstrates how automation, data processing, and cyber security concepts can be combined to support proactive threat detection and defensive security operations without using AI-based techniques.

Overall, the Threat Intelligence Aggregator (Non-AI) project highlights an efficient and reliable method for collecting and operationalizing threat intelligence. It provides hands-on experience in Blue Team security practices, IOC management, threat analysis, and cyber security automation, making it useful for academic learning as well as real-world security environments.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the development of my project "Threat Intelligence Aggregator (Non-AI)".

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Chapter 1

INTRODUCTION

The rapid growth of digital technologies has significantly increased the number and complexity of cyber threats faced by organizations worldwide. Cybercriminals continuously exploit vulnerabilities using malicious IP addresses, domains, URLs, malware, and phishing campaigns, making it essential for security teams to detect and respond to threats quickly. Security Operations Centers (SOCs) rely on cyber threat intelligence to identify Indicators of Compromise (IOCs) and strengthen an organization's defensive posture.

Threat intelligence is collected from multiple Open Source Intelligence (OSINT) feeds; however, manually gathering, validating, correlating, and analyzing these indicators is time-consuming and prone to human error. As the volume of threat data continues to grow, cyber security analysts require automated solutions that can efficiently process large datasets while maintaining accuracy and consistency.

The **Threat Intelligence Aggregator (Non-AI)** is a rule-based cyber security solution developed to automate the collection and processing of threat intelligence without using Artificial Intelligence or Machine Learning. The system retrieves threat indicators from multiple OSINT feeds, extracts and validates Indicators of Compromise (IOCs), normalizes heterogeneous data into a standard format, correlates repeated indicators, assigns severity levels, and generates defensive block lists and SOC-oriented reports.

The project has been implemented using Python and a modular architecture consisting of independent components responsible for feed loading, IOC parsing, normalization, correlation, severity classification, statistics generation, report generation, and block list creation. This modular design improves maintainability, scalability, and ease of future enhancement.

The primary objective of this project is to demonstrate how automation and rule-based intelligence processing can improve defensive cyber security operations. The generated outputs, including normalized IOC datasets, correlated reports, statistical summaries, and enforcement-ready block lists, provide practical support for SOC analysts and Blue Team operations. Overall, this project offers valuable hands-on experience in cyber security

automation, threat intelligence processing, and secure software development while demonstrating a practical approach to modern cyber defense.

Chapter 2

OBJECTIVES OF THE PROJECT

The primary objective of the Threat Intelligence Aggregator (Non-AI) project is to develop a rule-based cyber security solution that helps collect, process, and analyze threat intelligence data from multiple sources. The system focuses on improving threat detection capabilities by automating the handling of Indicators of Compromise (IOCs) without using Artificial Intelligence or Machine Learning techniques.

The major objectives of this project are:

- To collect threat intelligence data from multiple external Open-Source Intelligence (OSINT) feeds.
- To identify and extract different types of Indicators of Compromise (IOCs) such as malicious IP addresses, URLs, domains, and file hashes.
- To validate and normalize collected threat data into a common structured format for better analysis.
- To remove duplicate information and improve the quality and accuracy of threat intelligence data.
- To correlate indicators collected from different threat feeds and identify repeated malicious activities.
- To analyze the frequency of indicators and assign appropriate severity levels such as Low, Medium, and High.

- To prioritize critical threats and provide security analysts with actionable threat information.
- To generate block lists containing malicious indicators that can be used with security controls such as firewalls, endpoint protection systems, and web filters.
- To automate basic Security Operations Center (SOC) tasks and reduce manual investigation efforts.
- To produce structured reports and statistics that help security teams understand threat activity.

The project aims to demonstrate how cyber security automation and rule-based intelligence processing can improve defensive security operations. By organizing threat data into meaningful information, the system helps organizations respond faster to potential cyber threats and strengthen their overall security posture.

Chapter 3

SCOPE OF THE PROJECT

The scope of the Threat Intelligence Aggregator (Non-AI) project focuses on developing a rule-based cyber security solution for collecting, processing, and analyzing threat intelligence data. The project supports Blue Team and Security Operations Center (SOC) activities by providing structured threat information.

The project scope includes:

- Collection of threat intelligence data from multiple Open-Source Intelligence (OSINT) feeds.
- Extraction and identification of Indicators of Compromise (IOCs) such as malicious IP addresses, URLs, domains, and file hashes.
- Validation and normalization of collected threat data into a standard and structured format.
- Removal of duplicate or invalid indicators to improve data quality and accuracy.
- Correlation of threat indicators across multiple sources to identify repeated malicious activities.
- Classification of threats based on severity levels such as Low, Medium, and High.
- Prioritization of critical threats according to their occurrence and risk level.
- Generation of security block lists containing malicious indicators.

- Providing output that can be used with defensive security tools such as firewalls, end point protection systems, and web filters.
- Generation of threat intelligence reports and SOC-style statistics for security analysis.
- Automation of basic threat monitoring and analysis tasks to reduce manual investigation efforts.

The project is limited to defensive cyber security operations and does not include offensive activities. It does not use Artificial Intelligence or Machine Learning techniques and works completely on predefined rules and logical analysis.

Chapter 4

TOOLS & TECHNOLOGIES USED

Operating System

Kali Linux

Programming Language

Python 3

Python Libraries

- re – Regular expressions for IOC extraction
- requests – Feed retrieval
- json, csv – Data handling
- ipaddress – IP validation
- hashlib – Hash validation

Threat Intelligence Sources

- IPsum (malicious IP feeds)
- URLHaus (malicious URL feeds)

SYSTEM ARCHITECTURE DIAGRAM

THREAT INTELLIGENCE AGGREGATOR (NON-AI)

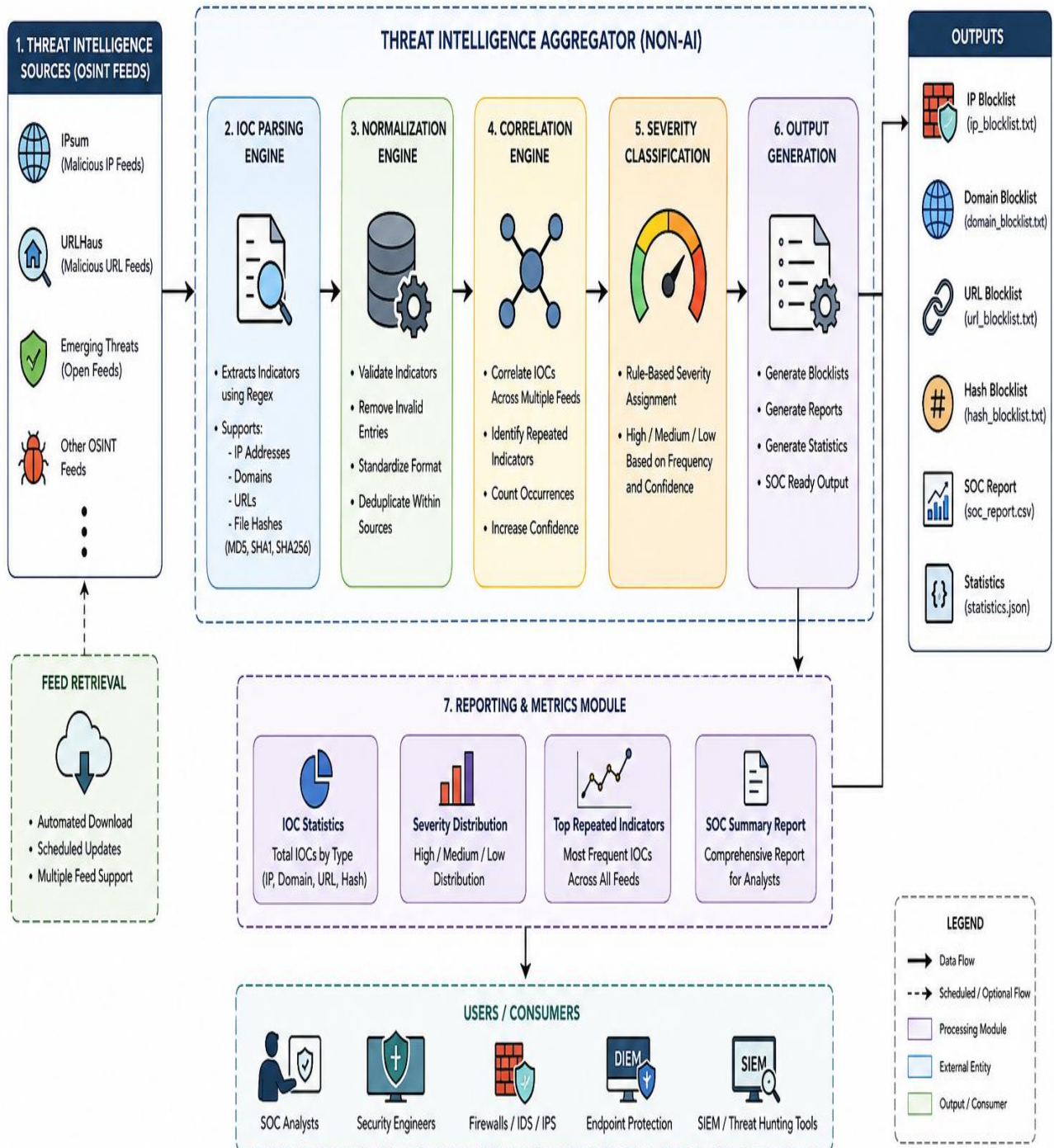


Figure 4.1 System Architecture of the Threat Intelligence Aggregator (Non-AI)

Chapter 5

FLOWCHART

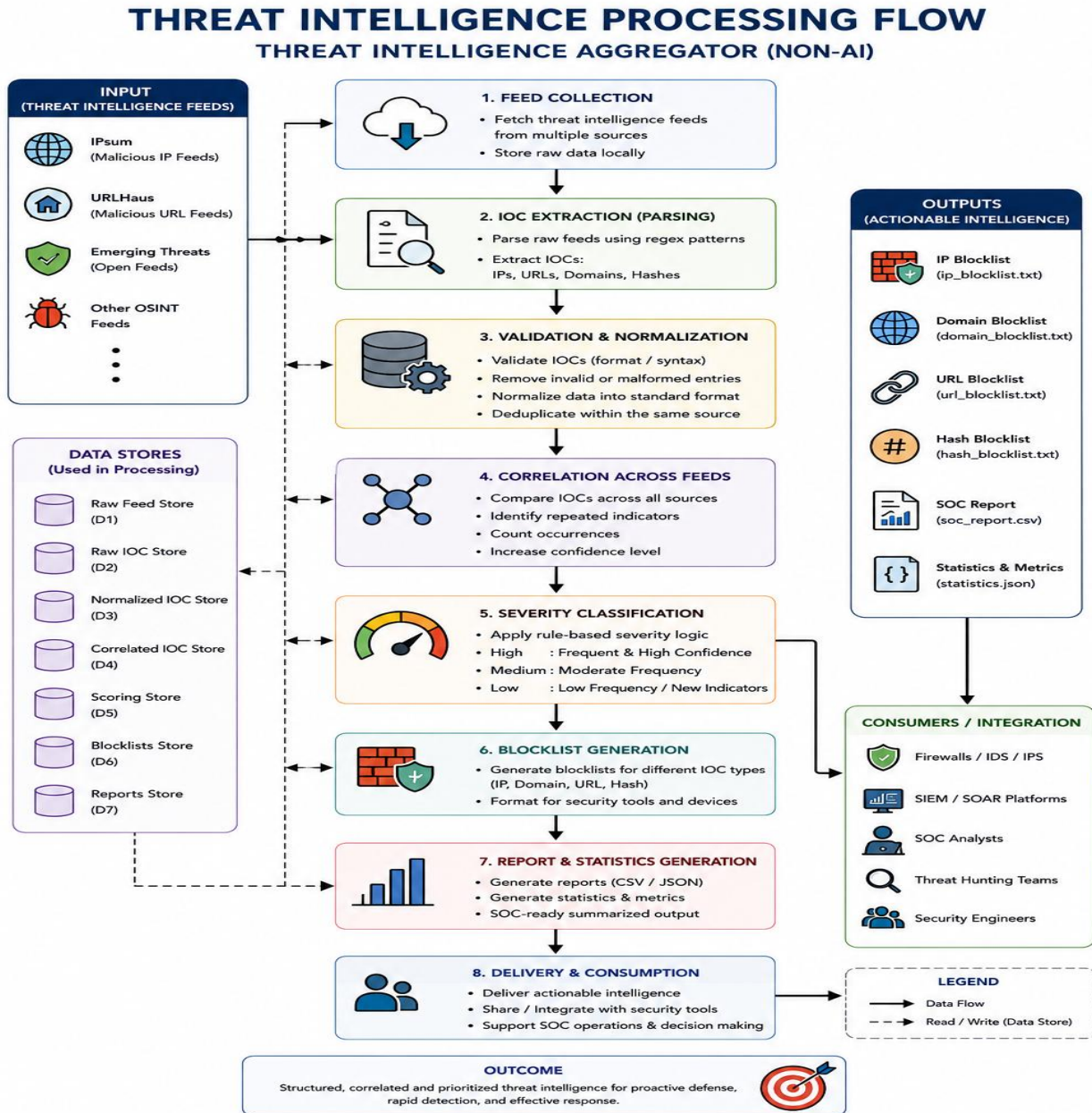


Figure 5.1 Threat Intelligence Processing Flow

WORKFLOW DIAGRAM

THREAT INTELLIGENCE AGGREGATOR (NON-AI)

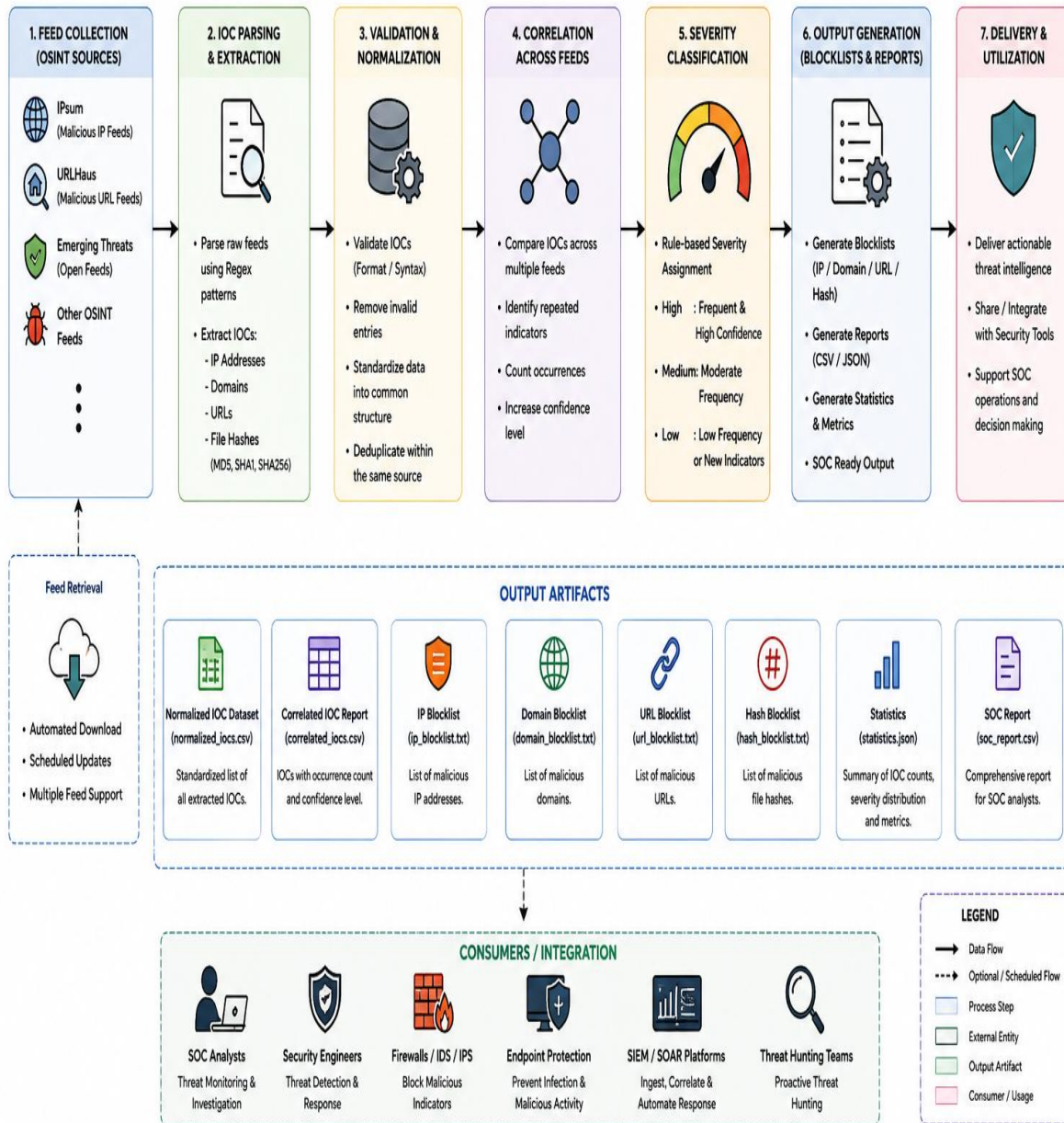


Figure 5.2 Overall Workflow of Threat Intelligence Aggregator

Chapter 6

IMPLEMENTATION DETAILS

The Threat Intelligence Aggregator (Non-AI) project is implemented using a modular Python-based architecture. The system is divided into multiple independent modules, where each module performs a specific task in the threat intelligence processing workflow. This modular approach improves maintainability, scalability, and ease of understanding.

1. ioc_parser.py

- Responsible for extracting Indicators of Compromise (IOCs) from raw threat intelligence feeds.
- Uses regular expressions to identify IOC types such as IP addresses, URLs, domains, and file hashes.

2. normalizer.py

- Performs validation and normalization of extracted indicators.
- Converts raw threat data into a standardized format for accurate processing and analysis.

3. correlator.py

- Correlates indicators collected from multiple threat intelligence feeds.
- Identifies repeated indicators and determines their importance based on occurrence frequency.

IOC CORRELATION PROCESS

THREAT INTELLIGENCE AGGREGATOR (NON-AI)

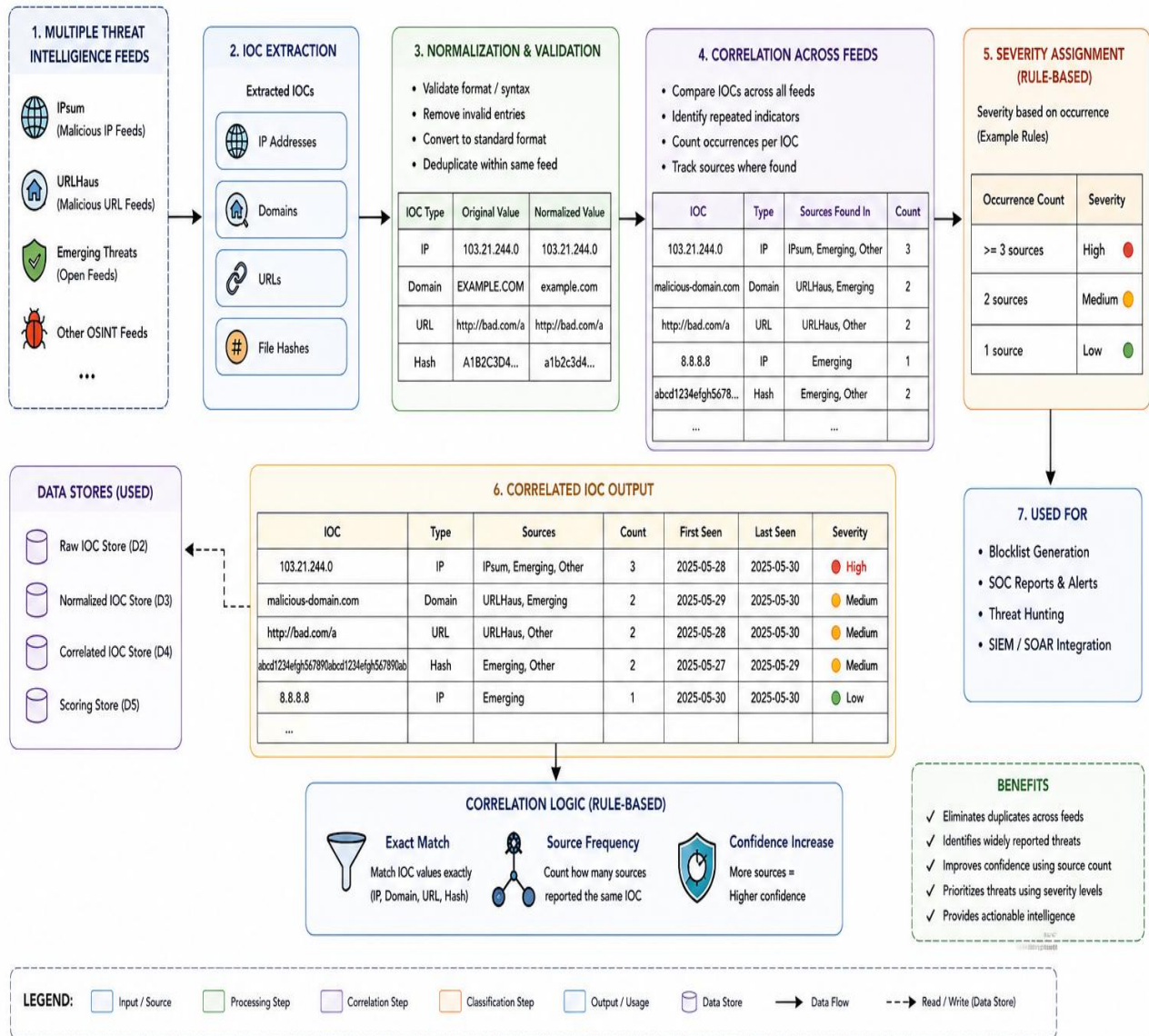


Figure 6.1 IOC Correlation Process

4. blocklist.py

- Generates enforcement-ready block lists containing high-risk indicators.

- These block lists can be used by security controls such as firewalls, end point security systems, and web filters.

5. stats.py

- Produces SOC-related metrics and summary information.
- Provides statistics such as total processed indicators and high-risk threat counts.

6. main.py

- Acts as the main controller of the complete workflow.
- Executes all modules in sequence from data collection to final report generation.

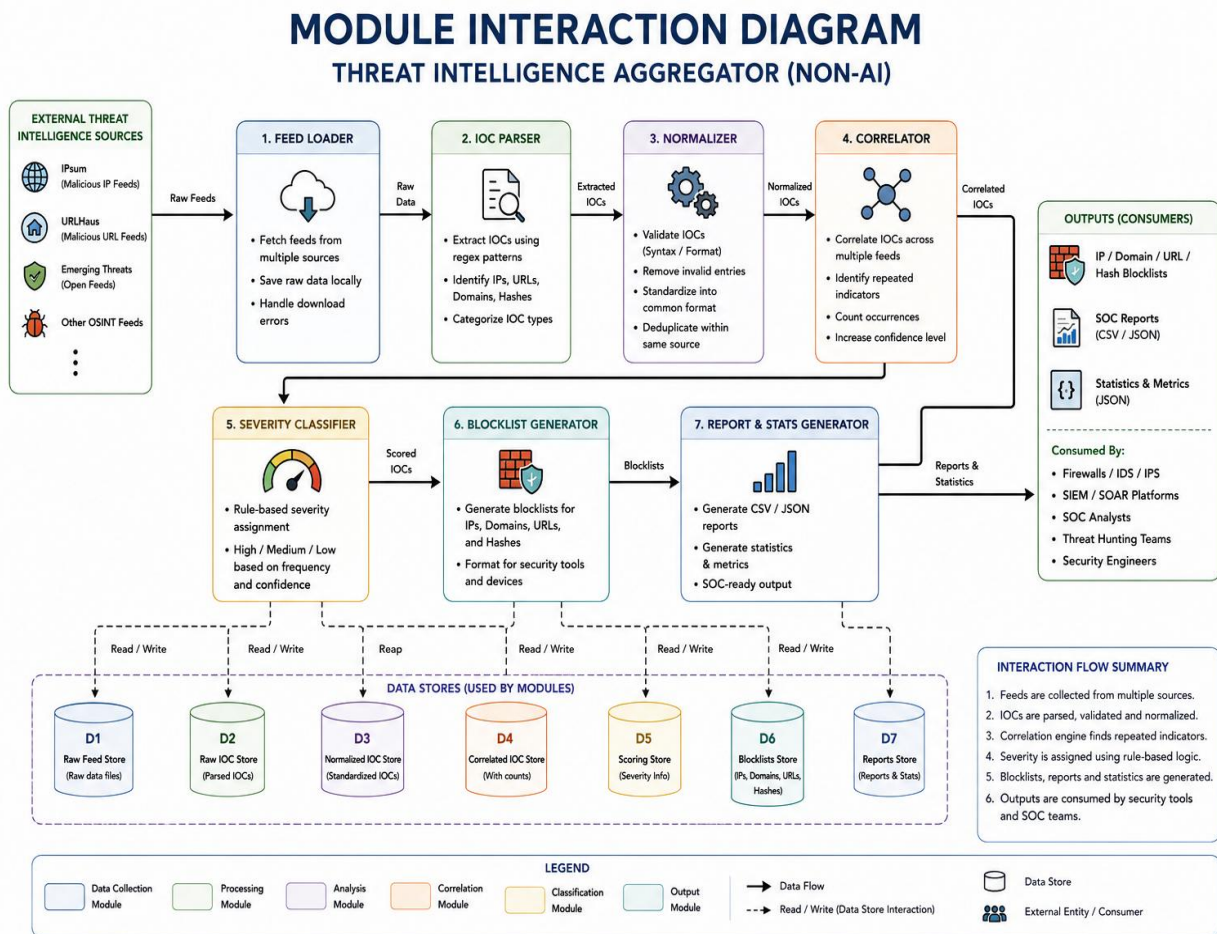


Figure 6.2 Module Interaction Diagram

DATA FLOW DIAGRAM (DFD)

THREAT INTELLIGENCE AGGREGATOR (NON-AI)

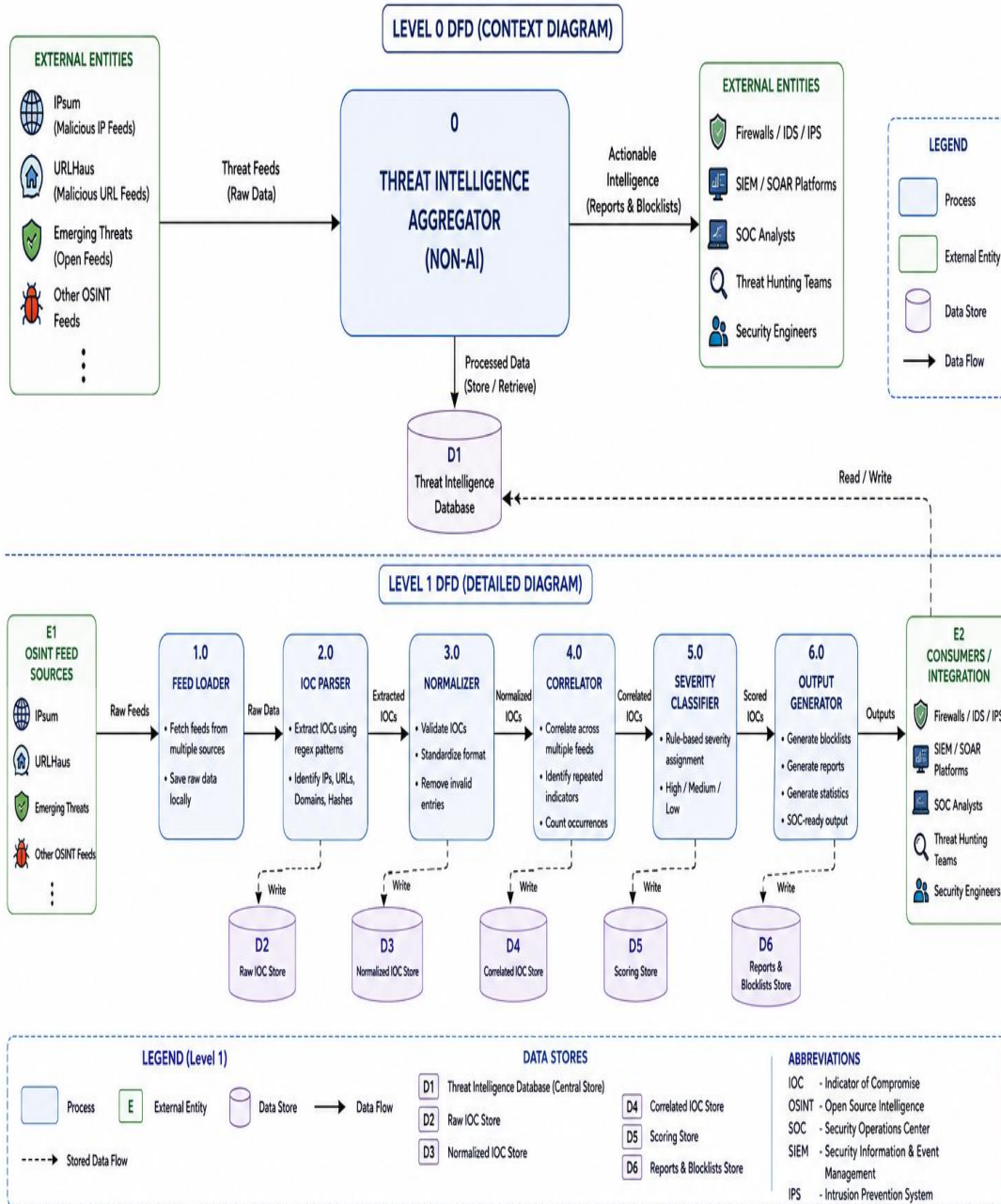


Figure 6.3 Data Flow Diagram (Level 0 & Level 1)

The complete system is executed on Kali Linux using Python 3 and tested with real Open-Source Intelligence (OSINT) threat data. The implementation demonstrates how automation and rule-based processing can support threat detection and defensive cyber security operations without using Artificial Intelligence or Machine Learning techniques.

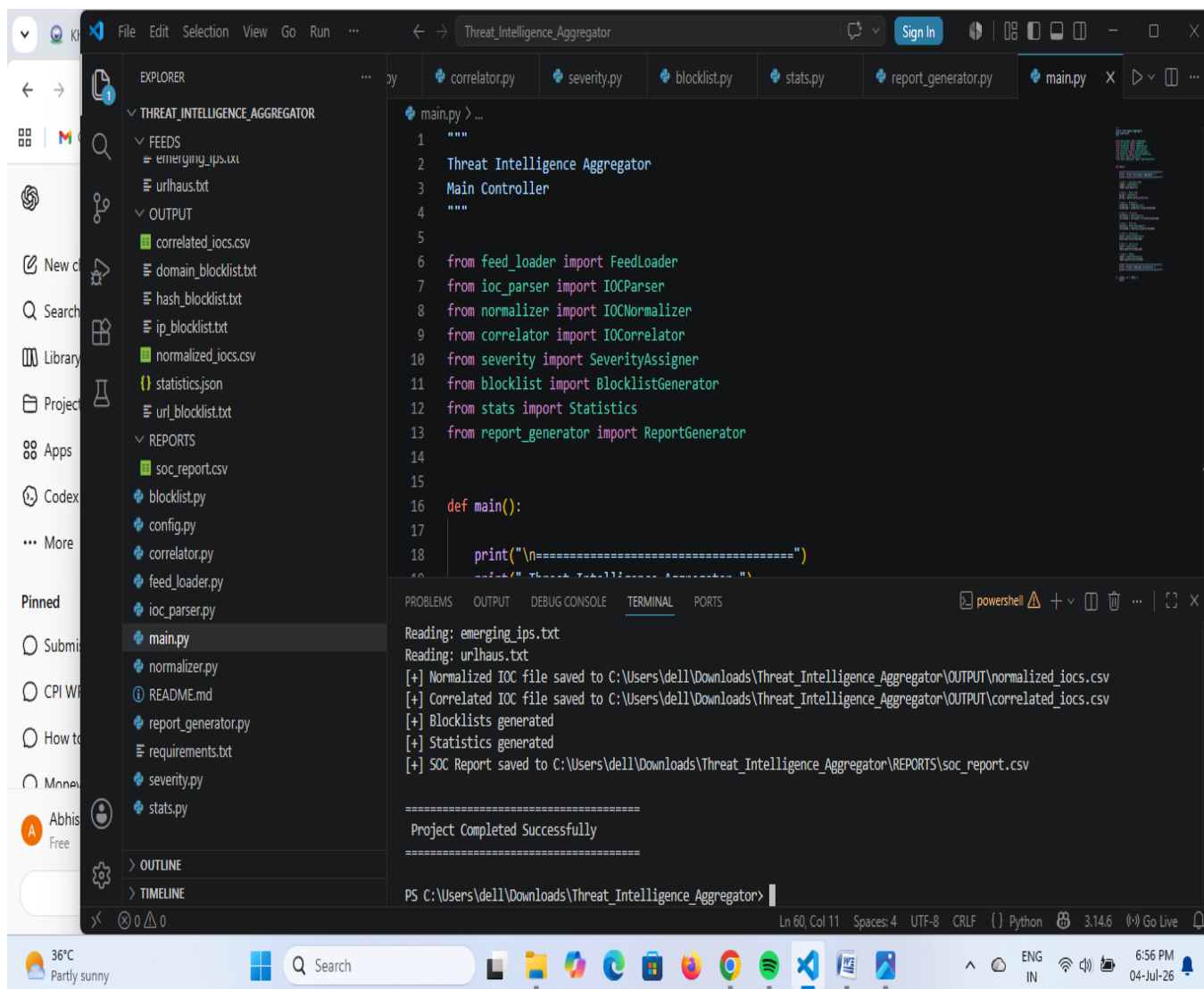


Figure 6.4 Successful Project Execution in Visual Studio Code

Chapter 7

OUTPUT GENERATED

Upon successful execution, the Threat Intelligence Aggregator generated multiple output files that support cyber threat analysis and Security Operations Center (SOC) operations. The system processed threat intelligence data collected from multiple Open Source Intelligence (OSINT) feeds and produced structured outputs for further investigation and defensive implementation.

The major outputs generated by the application are as follows:

- **Normalized IOC Dataset:** A standardized dataset containing extracted Indicators of Compromise (IOCs), including IP addresses, domain names, URLs, and file hashes. The normalization process converts data from different threat feeds into a consistent format, improving interoperability and simplifying analysis.
- **Correlated Indicators with Severity Levels:** The correlation module identifies duplicate indicators appearing across multiple threat intelligence sources and assigns severity levels (Low, Medium, or High) based on predefined occurrence rules.
- **Defensive Block lists:** The system automatically generates separate block list files for malicious IP addresses, domain names, URLs, and file hashes. These block lists can be integrated with firewalls, IDS, IPS, proxy servers, and SIEM platforms.
- **SOC-Oriented Threat Intelligence Statistics:** A statistical summary is generated in JSON format, providing the total number of processed indicators, IOC distribution by type, severity categorization, and overall threat intelligence metrics.

Example Outputs

- High-risk IP addresses appearing in multiple feeds.
- Consolidated malicious domain lists.

- Normalized IOC reports in CSV format.
- IP, Domain, URL, and Hash block lists.
- SOC report summarizing correlated indicators and severity.
- Statistics report containing processed IOC metrics.

The successful generation of these outputs demonstrates that the proposed Threat Intelligence Aggregator effectively automates the complete cyber threat intelligence processing workflow and provides actionable intelligence for cyber security analysts.

```

Threat_Intelligence_Aggregator
File Edit Selection View Go Run ...
EXPLORER
  THREAT_INTELLIGENCE_AGGREGATOR
    __pycache__
    FEEDS
      emerging_ips.txt
      urlhaus.txt
    OUTPUT
      correlated_iocs.csv
      domain_blocklist.txt
      hash_blocklist.txt
      ip_blocklist.txt
      normalized_iocs.csv
      statistics.json
      url_blocklist.txt
    REPORTS
      soc_report.csv
  severity.py
  blocklist.py
  stats.py
  report_generator.py
  main.py
  domain_blocklist.txt
  requirements.txt
  README.md
  config.py
  correlator.py
  feed_loader.py
  ioc_parser.py
  main.py
  normalizer.py
  report_generator.py
  requirements.txt
  severity.py
  stats.py
  OUTLINE
  TIMELINE

OUTPUT
  domain_blocklist.txt
  1 -07-26.rar
  2 -1.1-alpha.1.zip
  3 -1.21.4.jar
  4 -1.3.zip
  5 -1.4.zip
  6 -1.8.zip
  7 -12-1.0.zip
  8 -12-2.8.zip
  9 -160g.pdf
  10 -2.1.zip
  11 -2.8.zip
  12 -4022.pdf
  13 -91.pdf
  14 -Apex.exe
  15 -ApexLegends.exe
  16 -Assignment2-v3.3.zip
  17 -C00.exe

TERMINAL
  Reading: emerging_ips.txt
  Reading: urlhaus.txt
  [+] Normalized IOC file saved to C:\Users\dell\Downloads\Threat_Intelligence_Aggregator\OUTPUT\normalized_iocs.csv
  [+] Correlated IOC file saved to C:\Users\dell\Downloads\Threat_Intelligence_Aggregator\OUTPUT\correlated_iocs.csv
  [+] Blocklists generated
  [+] Statistics generated
  [+] SOC Report saved to C:\Users\dell\Downloads\Threat_Intelligence_Aggregator\REPORTS\soc_report.csv

  =====
  Project Completed Successfully
  =====

  PS C:\Users\dell\Downloads\Threat_Intelligence_Aggregator>

```

Figure 7.1: Successful Execution of the Threat Intelligence Aggregator in Visual Studio Code


```
1 type,Value
2 IP,1.10.209.143
3 IP,1.10.255.32
4 IP,1.171.42.227
5 IP,1.198.198.225
6 IP,1.2.185.103
7 IP,1.2.185.116
8 IP,1.2.4.1
9 IP,1.20.91.68
10 IP,1.22.174.37
11 IP,1.43.21.116
12 IP,1.58.106.149
13 IP,1.58.208.78
14 IP,1.61.209.11
15 IP,1.62.184.102
16 IP,1.62.184.87
17 IP,1.62.200.34
18 IP,1.62.241.196
19 IP,1.62.250.51
20 IP,100.23.75.120
21 IP,101.108.10.149
22 IP,101.108.103.139
23 IP,101.108.132.162
24 IP,101.108.146.151
25 IP,101.108.252.244
26 IP,101.108.3.225
27 IP,101.108.3.228
28 IP,101.108.66.29
29 IP,101.108.68.152
30 IP,101.109.203.143
31 IP,101.109.207.172
32 IP,101.109.81.159
```

Figure 7.2 – Normalized IOC Dataset

```
1 IOC_Type,IOC_Occurrences
2 ips,1.10.209.143,1
3 ips,1.10.255.32,1
4 ips,1.171.42.227,1
5 ips,1.198.198.225,1
6 ips,1.2.185.103,1
7 ips,1.2.185.116,1
8 ips,1.2.4.1,1
9 ips,1.20.91.68,1
10 ips,1.22.174.37,1
11 ips,1.43.21.116,1
12 ips,1.58.106.149,1
13 ips,1.58.208.78,1
14 ips,1.61.209.11,1
15 ips,1.62.184.102,1
16 ips,1.62.184.87,1
17 ips,1.62.200.34,1
18 ips,1.62.241.196,1
19 ips,1.62.250.51,1
20 ips,100.23.75.120,1
21 ips,101.108.10.149,1
22 ips,101.108.103.139,1
23 ips,101.108.132.162,1
24 ips,101.108.146.151,1
25 ips,101.108.252.244,1
26 ips,101.108.3.225,1
27 ips,101.108.3.228,1
28 ips,101.108.66.29,1
29 ips,101.108.68.152,1
30 ips,101.109.203.143,1
31 ips,101.109.207.172,1
32 ips,101.109.81.159,1
```

Figure 7.3 – Correlated IOC Dataset

The screenshot shows the Threat Intelligence Aggregator interface with the 'ur_blocklist.txt' file open. The file contains a list of URLs and IP addresses, including:

```
1 http://0000153.0000255.0000011.0000125/httpswww.tcs.comwhat-we-doindustriespublic-servicesolutiontcs-sovereignsecure-cloudness.php
2 http://0022a601.pphost.net/threatsim/doc/46cad0e0ca3bd6d9d3ce691ca2887b18abc80ac0e81799fb290cce104c8eb.doc
3 http://0022a601.pphost.net/threatsim/doc/774d0427cd607b1c09131cc27a68c9edd7cf01499d356bc1ef4a08e6fc322a.doc
4 http://0022a601.pphost.net/threatsim/exe/640.exe
5 http://0022a601.pphost.net/threatsim/exe/njpat.exe
6 http://0022a601.pphost.net/threatsim/exe/order.pdf.exe
7 http://0022a601.pphost.net/threatsim/exe/pdf.exe
8 http://0022a601.pphost.net/threatsim/exe/xerox01.pdf.exe
9 http://09c1d5c3-1a6e-4c05-8e4e-eff75c6b5dd6.filesusr.com/ugd/09c1d5_f29a3b493ad342199c678fc221f44af7.txt
10 http://09c1d5c3-1a6e-4c05-8e4e-eff75c6b5dd6.usrfiles.com/ugd/09c1d5_5408372e708a48629b728d0ea386361c.txt
11 http://05106.mira2x1.1187570e-aget-358671-chess-345msdn01/take.d11
12 http://1.10.289.143:17698/bin.sh
13 http://1.10.255.32:45960/bin.sh
14 http://1.10.255.32:45960/1
15 http://1.10.255.32:45960/1
16 http://1.2.185.103:35327/bin.sh
17 http://1.2.185.103:35327/1
18 http://1.2.185.103:35327/1
19 http://1.2.185.116:56553/bin.sh
20 http://1.20.91.68:35263/1
21 http://1.22.174.37:52272/bin.sh
22 http://1.22.174.37:52272/1
23 http://1.43.21.116:40105/bin.sh
24 http://1.43.21.116:40105/1
25 http://1.58.106.149:38782/bin.sh
26 http://1.58.106.149:38782/1
27 http://1.58.208.78:33017/bin.sh
28 http://1.61.209.11:44161/1
29 http://1.62.184.102:42183/bin.sh
30 http://1.62.184.102:42183/1
31 http://1.62.184.87:42183/bin.sh
32 http://1.62.184.87:42183/1
```

Figure 7.4 – Generated IP/Domain Blocklists

The screenshot shows the Threat Intelligence Aggregator interface with the 'soc_report.csv' file open. The file contains a list of IP addresses, their types, occurrences, and severity levels. The data is as follows:

IP	IOC_Type	IOC_Occurrences	Severity
1.10.200.143	1	1	LOW
1.10.255.32	1	1	LOW
1.171.42.222	1	1	LOW
1.198.108.225	1	1	LOW
1.2.185.103	1	1	LOW
1.2.185.116	1	1	LOW
1.2.4.1	1	1	LOW
1.20.91.68	1	1	LOW
1.22.174.37	1	1	LOW
1.43.21.116	1	1	LOW
1.58.106.149	1	1	LOW
1.58.208.78	1	1	LOW
1.61.209.11	1	1	LOW
1.62.184.102	1	1	LOW
1.62.184.87	1	1	LOW
1.62.200.34	1	1	LOW
1.62.241.100	1	1	LOW
1.62.250.51	1	1	LOW
100.23.75.120	1	1	LOW
101.108.10.149	1	1	LOW
101.108.103.139	1	1	LOW
101.108.132.162	1	1	LOW
101.108.146.151	1	1	LOW
101.108.252.244	1	1	LOW
101.108.3.225	1	1	LOW
101.108.3.228	1	1	LOW
101.108.66.29	1	1	LOW
101.108.68.152	1	1	LOW
101.109.283.143	1	1	LOW
101.109.287.172	1	1	LOW
101.109.81.159	1	1	LOW

Figure 7.5 – SOC Report

The screenshot shows the Threat Intelligence Aggregator interface with the 'statistics.json' file open. The file contains a JSON object with the following statistics:

```
1 {
2   "Total_IPs": 15589,
3   "Total_Domains": 21005,
4   "Total_URLs": 76511,
5   "Total_Mashes": 0,
6   "Total_IOCs": 113105
7 }
```

Figure 7.6 – Statistics Report

Chapter 8

Results and Analysis

The experimental evaluation of the **Threat Intelligence Aggregator** demonstrated that the proposed system successfully automated the complete cyber threat intelligence processing workflow. During execution, the application collected threat intelligence data from multiple Open Source Intelligence (OSINT) feeds, extracted Indicators of Compromise (IOCs), normalized heterogeneous data into a standardized format, correlated duplicate indicators, assigned severity levels, and generated comprehensive reports and defensive block lists.

The system efficiently processed **more than 113,000 Indicators of Compromise (IOCs)**, including IP addresses, domain names, and URLs obtained from multiple threat intelligence sources. The correlation engine identified indicators that appeared across multiple feeds, thereby increasing the confidence level associated with those indicators. Based on the predefined severity classification rules, indicators occurring repeatedly were assigned **High Severity**, while less frequently occurring indicators were categorized as **Medium** or **Low Severity**.

The generated block lists consisted of malicious IP addresses, domains, URLs, and file hashes that can be directly integrated into firewalls, Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), proxy servers, and Security Information and Event Management (SIEM) platforms. Additionally, the application generated SOC-oriented reports and statistical summaries that provide cyber security analysts with a consolidated view of the processed threat intelligence.

The successful execution of all system modules without runtime errors confirms the reliability, accuracy, and efficiency of the proposed implementation. Compared with manual threat intelligence analysis, the developed system significantly reduces processing time, minimizes human intervention, improves consistency, and enables faster identification of potentially malicious indicators.

The results demonstrate that the Threat Intelligence Aggregator provides practical support for Security Operations Center (SOC) activities, including proactive threat monitoring, automated threat prioritization, defensive block list generation, and improved decision-

making. These capabilities make the proposed system a valuable tool for academic research, cyber security training, and small to medium-scale organizational security operations.

Key Observations

- Successfully processed 113,376+ Indicators of Compromise (IOCs).
- Automatically identified repeated indicators across different threat intelligence sources.
- Classified threats into Low, Medium, and High severity levels.
- Generated SOC reports, normalized IOC datasets, and statistical summaries.
- Created IP, Domain, URL, and Hash block lists for defensive deployment.
- Improved efficiency, reduced manual effort, and enhanced threat prioritization.

Chapter 9

Learning Outcomes

The development of the **Threat Intelligence Aggregator** provided valuable practical experience in cyber security, software development, and threat intelligence automation. Throughout the project lifecycle, several technical and analytical skills were acquired while designing, implementing, testing, and evaluating the proposed system.

The key learning outcomes include:

- **Understanding Threat Intelligence Feeds:** Learned how Open Source Intelligence (OSINT) feeds provide valuable information about emerging cyber threats and malicious infrastructure.
- **IOC Extraction and Validation:** Gained practical experience in identifying, extracting, validating, and organizing Indicators of Compromise (IOCs) such as IP addresses, domains, URLs, and file hashes.
- **Data Normalization Techniques:** Understood methods for converting heterogeneous threat data into a standardized structure for efficient processing and analysis.
- **Threat Correlation Logic:** Developed knowledge of correlating repeated indicators across multiple intelligence sources to improve confidence and prioritize threats.
- **SOC Workflow Automation:** Learned how automation can streamline Security Operations Center (SOC) workflows by reducing manual effort and generating structured reports and block lists.
- **Defensive Cyber security Practices:** Enhanced understanding of proactive cyber defense through block list generation, threat prioritization, and intelligence-driven security operations.

Overall, this project strengthened practical skills in Python programming, cyber security automation, threat intelligence processing, and secure software development while providing hands-on exposure to real-world SOC concepts.

Chapter 10

Conclusion

The **Threat Intelligence Aggregator (Non-AI)** project successfully demonstrates how cyber threat intelligence can be collected, processed, correlated, and transformed into actionable security information without relying on Artificial Intelligence or Machine Learning techniques. By utilizing a rule-based approach, the system automates the collection of Indicators of Compromise (IOCs) from multiple Open Source Intelligence (OSINT) feeds, normalizes heterogeneous data, identifies repeated indicators, assigns severity levels, and generates block lists and Security Operations Center (SOC) reports.

The implementation confirms that automation alone can significantly improve the efficiency of threat intelligence processing while reducing manual effort and ensuring consistent results. The generated outputs, including normalized IOC datasets, correlated threat reports, block lists, and statistical summaries, provide valuable support for cyber security analysts in identifying and prioritizing potential threats.

The project closely aligns with modern **Blue Team** and **Security Operations Center (SOC)** workflows by supporting proactive threat monitoring, defensive security operations, and intelligence-driven decision making. Furthermore, the modular architecture makes the system easy to maintain, extend, and integrate with additional threat feeds or enterprise security solutions in the future.

Overall, the Threat Intelligence Aggregator serves as a practical and cost-effective cyber security solution while providing valuable hands-on experience in threat intelligence processing, SOC automation, and defensive cyber security practices. The project successfully achieves its objectives and establishes a strong foundation for future

enhancements such as real-time intelligence integration, SIEM/SOAR connectivity, advanced visualization dashboards, and AI-assisted threat prioritization.

Chapter 11

Future Enhancements

Although the proposed **Threat Intelligence Aggregator** successfully automates the collection, processing, and reporting of cyber threat intelligence, several enhancements can further improve its scalability, interoperability, automation capabilities, and practical applicability within enterprise cyber security environments.

1. Support for STIX/TAXII Threat Intelligence Feeds

Future versions can integrate industry-standard threat intelligence formats such as **STIX** (Structured Threat Information Expression) and **TAXII** (Trusted Automated Exchange of Intelligence Information). This would enable seamless sharing of threat intelligence with commercial and open-source Cyber Threat Intelligence (CTI) platforms.

2. SIEM-Ready Output Formats

The generated reports and blocklists can be enhanced for direct integration with enterprise Security Information and Event Management (SIEM) platforms such as Splunk, IBM QRadar, Microsoft Sentinel, Elastic Security, and Wazuh, enabling automated threat correlation and monitoring.

3. Scheduled Feed Updates

Future implementations can automatically retrieve the latest threat intelligence feeds using Cron Jobs, Windows Task Scheduler, or other scheduling mechanisms, ensuring continuous updates without manual execution.

4. Logging and Alerting Mechanism

A comprehensive logging and alerting module can maintain execution logs and notify security analysts through email, Slack, Microsoft Teams, or web hooks whenever high-severity threats or processing errors are detected.

5. Interactive Dashboard Visualization

A web-based dashboard developed using Flask, Django, or React can provide interactive

charts, IOC statistics, severity trends, searchable indicators, and real-time monitoring, improving usability for Security Operations Center (SOC) analysts.

Overall Future Scope

These enhancements would transform the Threat Intelligence Aggregator into a comprehensive cyber threat intelligence platform capable of supporting real-time security operations, enterprise-scale threat monitoring, and advanced defensive cyber security practices.

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